# Anish Ray

*PhD Student in Mathematics, University of Houston*
*3551 Cullen Blvd., Houston, TX 77204*

## PERSONAL DETAILS

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|---|---|
| *E-mail* | `aray9@uh.edu` |
| *Date of Birth* | May 27, 2000 |
| *Nationality* | Indian |
| *Website* | `https://math.uh.edu/~aray8` |

## EDUCATION

**Ph.D. in Mathematics** **August 2025 – Present**
*University of Houston, USA*
Current GPA: 3.778

**M2 Mathématiques et Applications, Spécialité Analyse, Arithmétique, Géométrie** **2024 – 2025**
*Institut de mathématique d'Orsay, Université Paris-Saclay, France*
Overall grade: 12.01 (*Assez Bien*)

**M.Sc. in Mathematics** **2020 – 2023**
*Universität Münster, Germany*
Overall grade: 1.8 (*Gut*)

**B.Sc. (Hons.) in Mathematics and Computing** **2017 – 2020**
*Institute of Mathematics and Applications, Bhubaneswar (IMA), affiliated with Utkal University, Bhubaneswar, Odisha, India*
Grade: First Class with Distinction

## RESEARCH INTERESTS

Analytic Number Theory, $L$-functions, Spectral Theory of Automorphic Forms, Automorphic Representations, Arithmetic Statistics, Equidistribution Problems, and Homogeneous Dynamics.

## PREPRINTS

1. *On the Constants of the Lang-Trotter Conjecture for CM Elliptic Curves.*
`arXiv:2309.09938 [math.NT]`

## RESEARCH AND READING PROJECTS

**PIMS – First Year Interest Groups** **October 2025 – May 2026**
*Group Reading Project on Primes and the Theory of the Riemann Zeta Function, Pacific Institute for the Mathematical Sciences*
Supervisor: Dr. Nicol Leong, Postdoctoral Fellow, University of Lethbridge

**M2 Thesis** **December 2024 – June 2025**
*On Subconvexity of $L$-functions on $U(n) \times U(n+1)$ in the Depth Aspect*
Supervisor: Prof. Farrell Brumley, IMJ-PRG, Sorbonne Université, France Grade: 14.5/20 (*Bien*)

**Master's Thesis** **September 2022 – March 2023**

*On the Constants of the Lang-Trotter Conjecture for CM Elliptic Curves*

Supervisor: Prof. Dr. Christopher Deninger, Mathematisches Institut, Universität Münster, Germany Grade: 1.0 (Highest Grade)

**Research Project** **April 2023 – April 2024**

*Asymptotics of the Number of Solutions of Forms Having Prime Coordinates via the Hardy-Littlewood Circle Method*

Supervisor: Prof. Julia Brandes, Chalmers University of Technology, Gothenburg, Sweden

**Research Project** **November 2022 – October 2023**

*Developed a New Large Sieve Inequality by Introducing Suitable Weight Functions*

Supervisor: Prof. Dr. Jan-Christoph Schlage-Puchta, Universität Rostock, Germany

**Bachelor's Project** **February 2020 – May 2020**

*Some Problems on Elementary Number Theory*

Supervisor: Dr. Sudhansu Sekhar Rout (currently Asst. Prof., NIT Calicut, Kerala, India)

**Summer Project (Visiting Summer Research Programme)** **June 2019 – July 2019**

*Reading Project on the Spectral Theory of Unbounded Operators*

Supervisor: Dr. Shyam Ghoshal, TIFR Centre for Applicable Mathematics (CAM), Bengaluru, India

## ACADEMIC AWARDS

**AMS–NSF ICM 2026 Graduate Student Travel Grant** **2026**

*Awarded by the American Mathematical Society supporting travel for poster presentation at ICM 2026 in Philadelphia.*

**Presidential Fellowship** **2025 – 2027**

*Awarded by the University of Houston to recruit doctoral students with outstanding credentials. The fellowship provides $2,000 in addition to the Graduate Assistantship for the first two years of the PhD programme.*

**Graduate Tuition Fellowship** **2025 – Present**

*Competitive fellowship awarded by the University of Houston Graduate School covering in-state tuition and mandatory fees for PhD students.*

**M2 Internship Allowance** **2025**

*Awarded by the Institut de Mathématiques de Jussieu-Paris Rive Gauche (IMJ-PRG) to support thesis completion.*

Monthly allowance of €660 for three months.

**IDEX Excellence Scholarship** **2024 – 2025**

*Awarded by the Université Paris-Saclay to only 80 international students pursuing a master's degree.*

Total of €10,000 for ten months, plus €850 in travel expenses.

**Graduation Scholarship** **2021 – 2022**

*Awarded by the International Office, WWU Münster.*

Total of €1,500 for three months.

**Merit Scholarship** **2015**

*One-time merit scholarship of INR 5,000 awarded by the Kendriya Vidyalaya Sangathan (KVS) for securing 10 CGPA in AISSE 2015.*

## CONFERENCE AND SEMINAR TALKS

**International Congress of Mathematicians** **2026**

*Philadelphia*

Poster presentation: *"A Computational Comparison of Lang-Trotter and Hardy-Littlewood Constants for CM Elliptic Curves"* at the ICM in July 2026.

**Graduate Student Paper Presentation** **2026**
*Organized by the UH-AMS Graduate Chapter, University of Houston*

**Probabilistic Number Theory: Law of Large Numbers and CLT in Number Theory** **2022**
*Universität Münster*

**Linear Algebraic Groups: Connected Solvable Groups and Maximal Tori** **2022**
*Universität Münster*

**Elliptic Curves, Tate Module and the Weil Pairing** **2022**
*Universität Münster*

**Class Field Theory** **2021**
*Universität Münster*

**Zeta and $L$-functions in Algebraic Number Theory** **2021**
*Universität Münster*

## CONFERENCES, SUMMER SCHOOLS, AND WORKSHOPS ATTENDED

**Recent Perspectives on Moments of L-functions** **2026**
*University of Texas at Dallas*

**Workshop on Automorphic Forms and Number Theory** **2025**
*IMJ-PRG, Sorbonne University, Paris*

**Summer School on Inclusive Paths in Explicit Number Theory** **2023**
*Banff International Research Station (BIRS) at the University of British Columbia – Okanagan, Canada*

**Summer School on Automorphic Forms** **2022**
*Alfréd Rényi Institute of Mathematics, Budapest*

## TEACHING

**Teaching Assistant, University of Houston** **Spring 2026**
*MATH 2451 – Accelerated Calculus II*

**Teaching Assistant, University of Houston** **Fall 2025, Spring 2026**
*MATH 2305 – Discrete Mathematics*

**Associate Faculty, Cheenta Ganit Kendra (`https://cheenta.com/`)** **2017 – 2018**
*Online tutoring platform for aspirants of Mathematics Olympiads and national-level entrance examinations.*
Conducted lectures, problem-solving sessions, and created and graded assignments, quizzes, and examinations.

## ACADEMIC SERVICE AND OUTREACH

**Proofreader, University of Houston Mathematics Contest** **2026**
*The University of Houston hosts a major annual Mathematics Contest for high school and middle school students, featuring individual and team examinations across multiple tracks.*
`https://mathcontest.uh.edu/`

**Head of Technical Affairs and Problem Selection Committee** **2018**
*Scholastic Aptitude Test in Mathematics (SATM), conducted by the student association of the Institute of Mathematics and Applications, Bhubaneswar, as part of its annual techno-cultural fest. The event focuses on spreading awareness about higher education in mathematics and organises training camps for selected high school students in Odisha, India.*
`https://sites.google.com/view/satm-ima`

## SKILLS

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| *Programming* | C, C++, Python, JavaScript, Java, Visual Basic |
| *Typesetting* | LaTeX |
| *Markup* | HTML |
| *Stylesheet* | CSS |
| *Office Software* | MS Word, MS Excel, MS PowerPoint, MS Access |
| *Languages* | Bengali (mother tongue)<br>English (professional)<br>German (proficient, A2)<br>French (proficient, A1)<br>Hindi (native)<br>Sanskrit (intermediate) |